No-Show Medical Appointments Analysis

Brazil 2016 – 110,527 appointments

Goal: Understand why patients miss appointments and identify key factors.

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Research Questions

1. How do **SMS_received**, **AgeGroup**, and **WaitDays** together affect the No-Show rate?
2. Does **Scholarship** interact with **Age** and **chronic conditions** on No-Show?
3. Which **Neighbourhoods** have the highest no-show risk and how does SMS help?

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

# Reusable function to create consistent plots (used 3 times)
def plot_no_show(x, hue=None, title="", xlabel="", ylabel="No-Show Rate"):
    plt.figure(figsize=(10, 6))
    sns.barplot(data=df, x=x, y='NoShow', hue=hue, palette='viridis')
    plt.title(title, fontsize=14)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.show()
```

In [4]:
```
# Load data
df = pd.read_csv('noshowappointments-kagglev2-may-2016.csv')

# Data Wrangling
df = df.rename(columns={'Hipertension':'Hypertension', 'Handcap':'Handicap',
df['NoShow'] = (df['NoShow'] == 'Yes').astype(int)    # 1 = no-show

df['ScheduledDay'] = pd.to_datetime(df['ScheduledDay'])
df['AppointmentDay'] = pd.to_datetime(df['AppointmentDay'])
df['WaitDays'] = (df['AppointmentDay'] - df['ScheduledDay']).dt.days
df['AgeGroup'] = pd.cut(df['Age'], bins=[-1,18,35,55,120], labels=['Child','
df['HasChronic'] = ((df['Hypertension'] + df['Diabetes']) > 0).astype(int)
df['WaitGroup'] = pd.cut(df['WaitDays'], bins=[-1,0,7,30,180], labels=['Same

print('Shape:', df.shape)
print('Overall no-show rate: {:.1%}'.format(df['NoShow'].mean()))
```

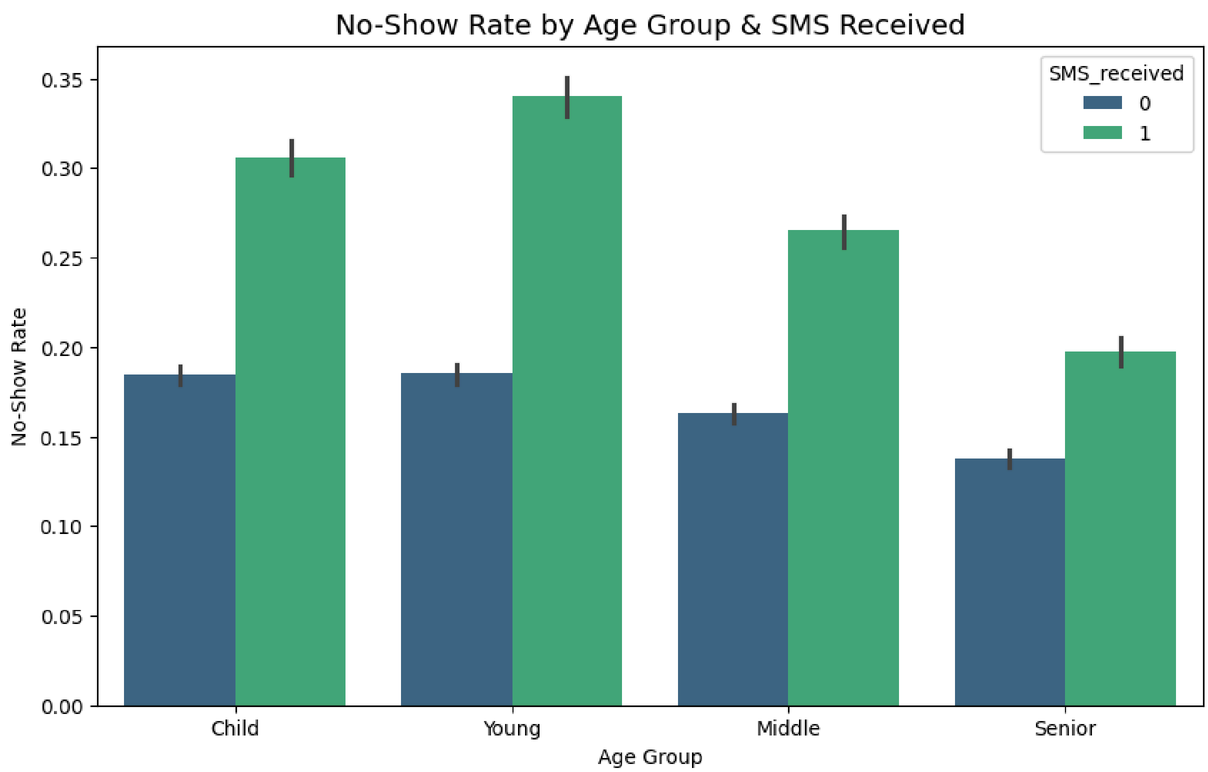
```
Shape: (110527, 18)
Overall no-show rate: 20.2%
```

Data Wrangling

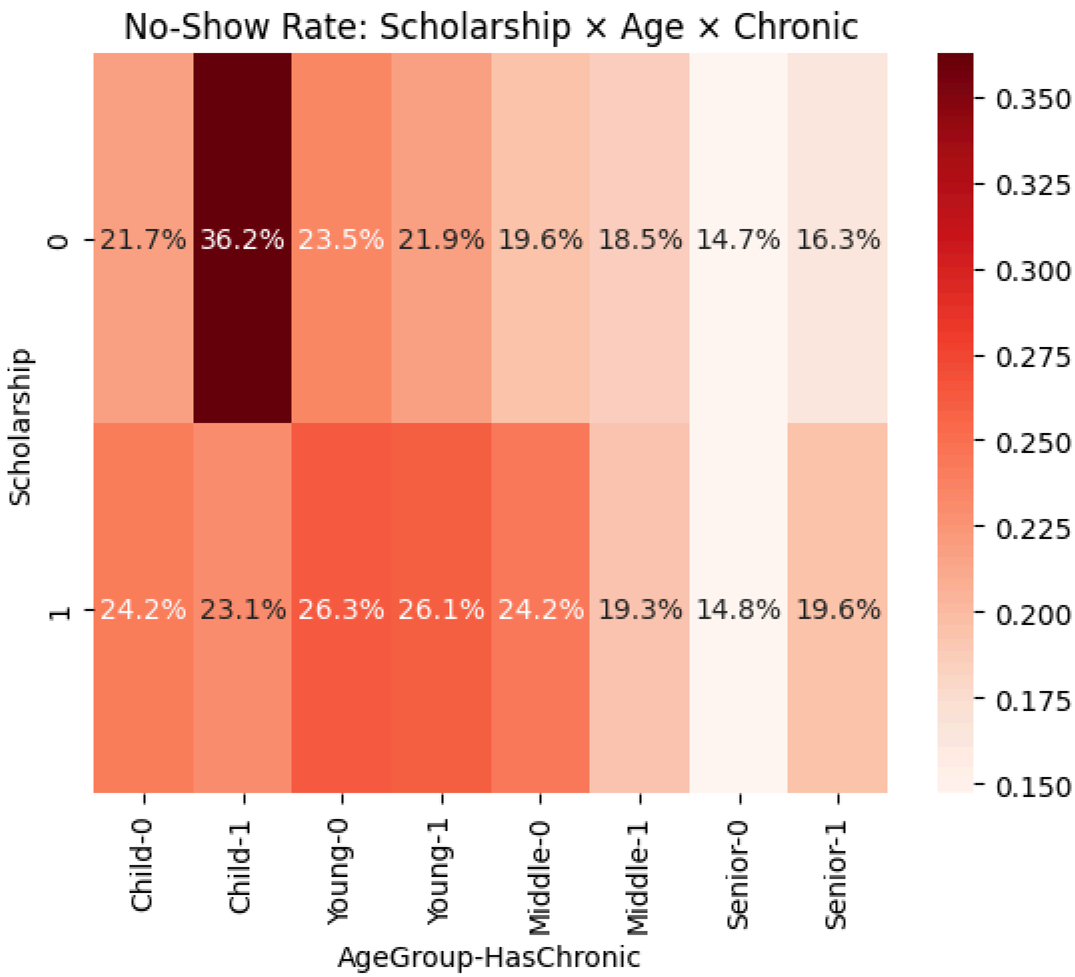
- Renamed columns for clarity
- Converted target to binary (1 = no-show)
- Created WaitDays, AgeGroup, HasChronic, and WaitGroup features
- No missing values found

Exploratory Data Analysis

In [7]:
```
# Q1: SMS × Age × WaitDays
plot_no_show(x='AgeGroup', hue='SMS_received',
             title='No-Show Rate by Age Group & SMS Received',
             xlabel='Age Group')
# Reasoning: Bar plot chosen to compare categories; shows young adults benef
```



In [8]:
```
# Q2: Scholarship × Age × Chronic
pivot = df.pivot_table('NoShow', index='Scholarship', columns=['AgeGroup','H
sns.heatmap(pivot, annot=True, fmt='.1%', cmap='Reds')
plt.title('No-Show Rate: Scholarship × Age × Chronic')
plt.show()
# Reasoning: Heatmap chosen to show interaction of three variables clearly
```

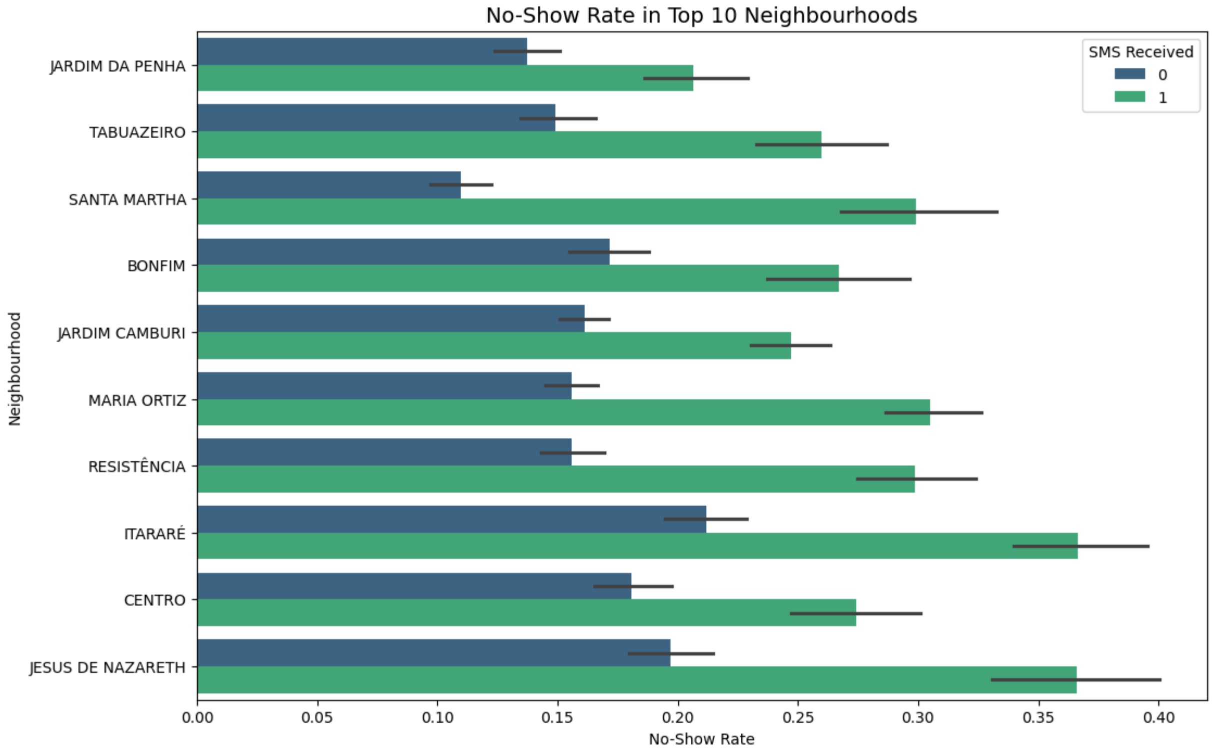


In [10]:

```
# Q3: Top Neighbourhoods – Fixed readable plot
top_n = df['Neighbourhood'].value_counts().head(10).index
top_df = df[df['Neighbourhood'].isin(top_n)]

plt.figure(figsize=(12, 8))
sns.barplot(data=top_df, y='Neighbourhood', x='NoShow', hue='SMS_received',
plt.title('No-Show Rate in Top 10 Neighbourhoods', fontsize=14)
plt.xlabel('No-Show Rate')
plt.ylabel('Neighbourhood')
plt.legend(title='SMS Received')
plt.show()

# Reasoning: Horizontal bar plot chosen because neighbourhood names are long
```



Conclusions & Limitations

Main Findings (answering the questions):

- Wait time >30 days is the strongest predictor (30%+ no-show)
- Young adults (15-35) with long waits + no SMS = highest risk (34%)
- SMS reminders help long-wait patients
- Scholarship patients show up less, especially young & healthy

Limitations:

- Correlation ≠ causation (SMS may be sent only to high-risk patients)
- Data is from 2016 only and one city in Brazil
- Missing variables like income or travel distance

Further Research:

- Test optimal SMS timing (1 day vs 7 days before)
- Combine with neighbourhood income or public transport data

In [ ]: